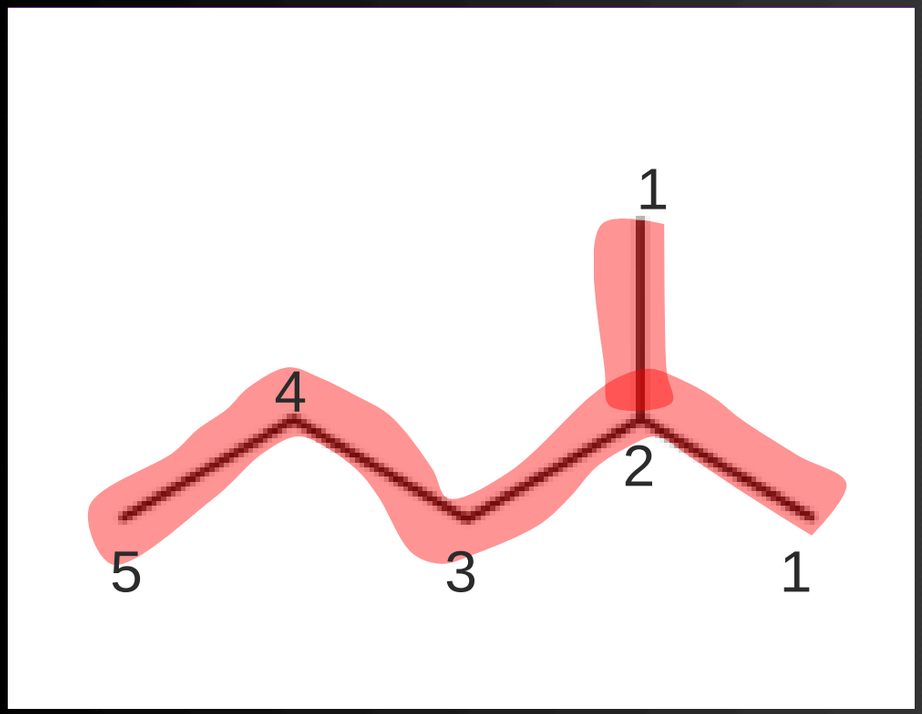
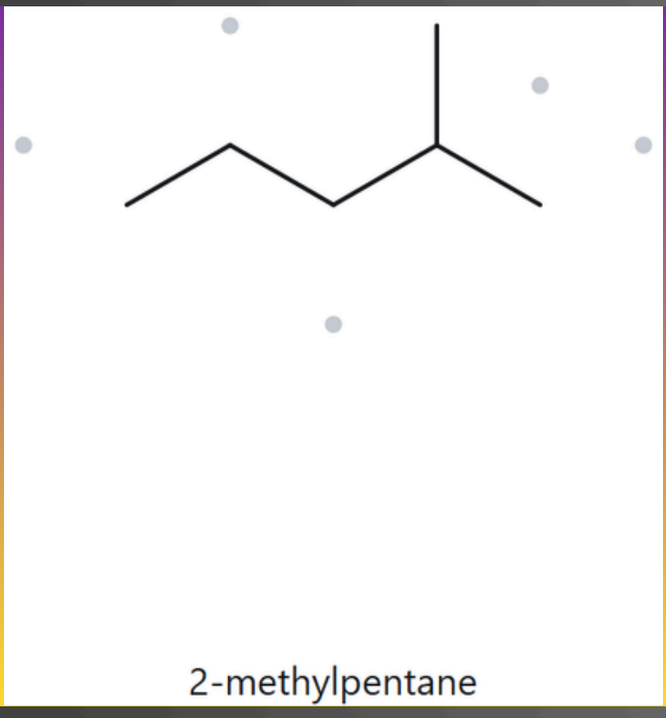
2-METHYLPENTANE A01-FOUNDATION C_6H_{14}



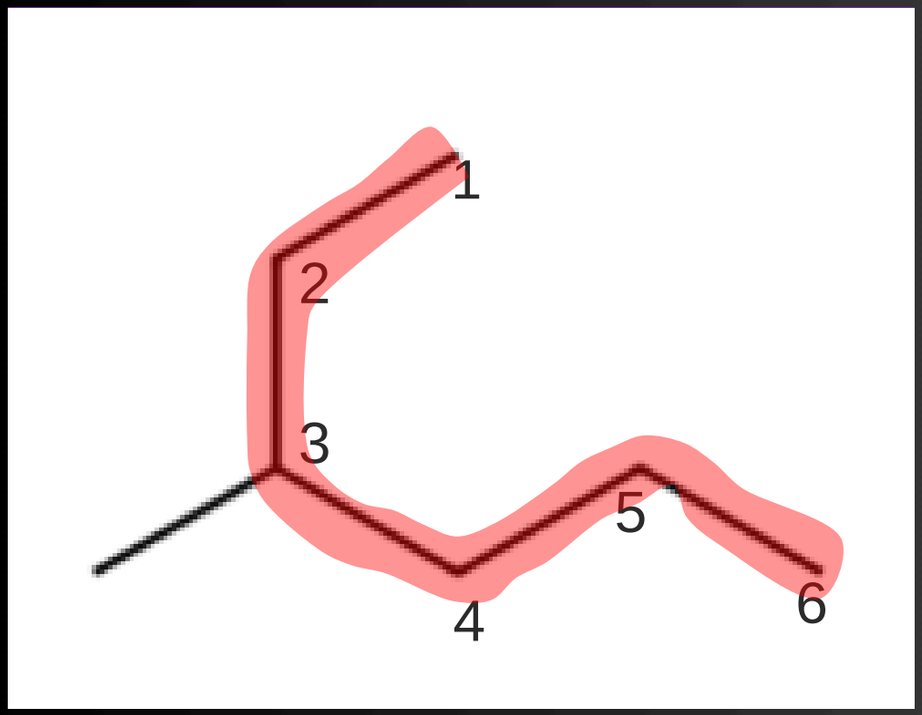
Parent: longest chain= 5 C, Pentane. Several 5-C paths exist but each carries 1 methyl group so they are the same.
Numbering: Starting from the right from any of the two branches gives the methyl group the lowest locant compared to starting from the left.
Substituents: One methyl group on C2. 2-methyl

2-methylpentane



LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

3-METHYLHEXANE A02-FOUNDATION C_7H_{16}

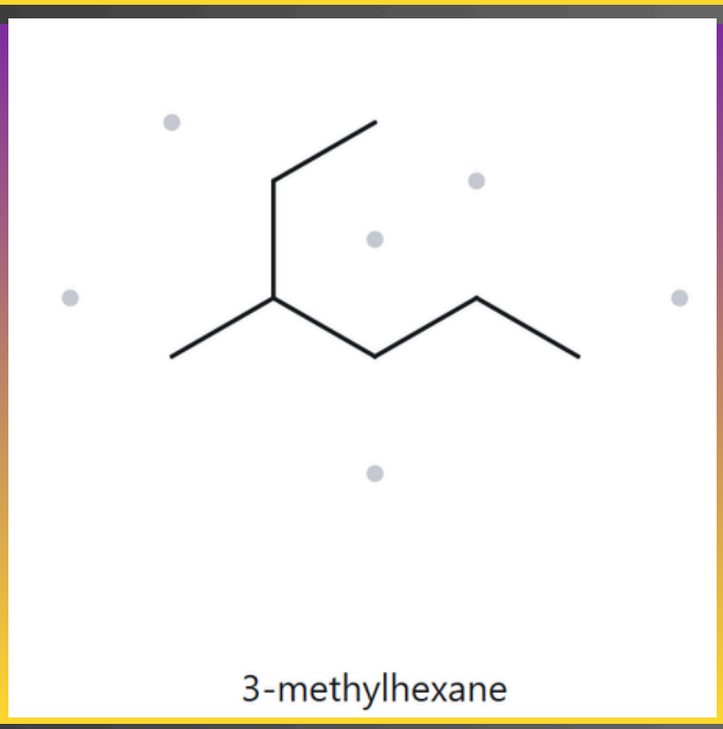


Parent: The longest chain is 6 C, so the parent is hexane. It runs from the top CH_3 up through the CH_2 to the branching carbon, then along the bottom row to the right. The 5-C bottom row is not the longest path.

Numbering: Number from the top CH_3. The methyl lands on C3. Numbering from the right-hand CH_3 would put it on C4, so 3 < 4 and the bottom end is correct.

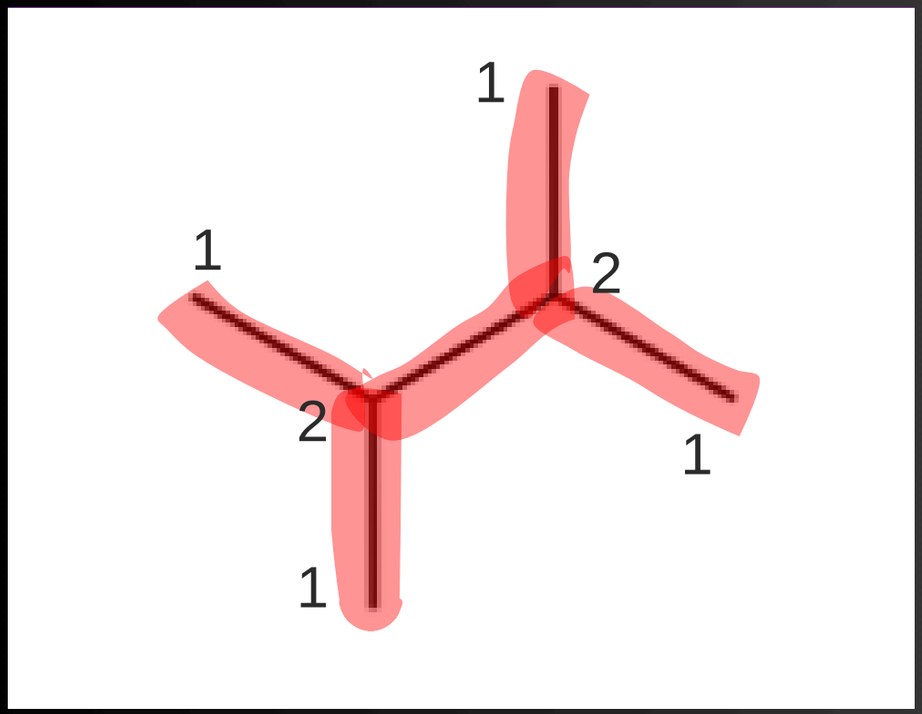
Substituents: One methyl group on C3. 3-methyl

3-methylhexane



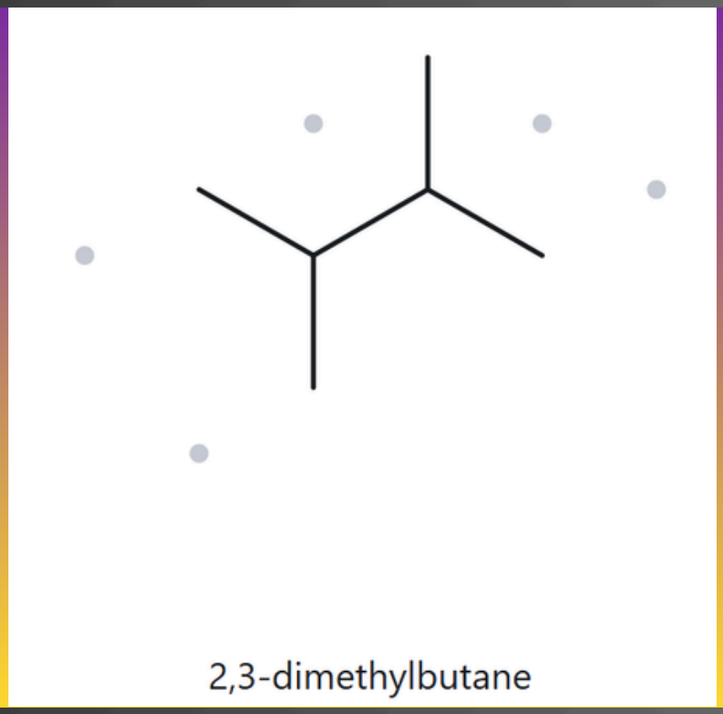
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

2,3-DIMETHYLBUTANE AO3-FOUNDATION C_6H_{14}



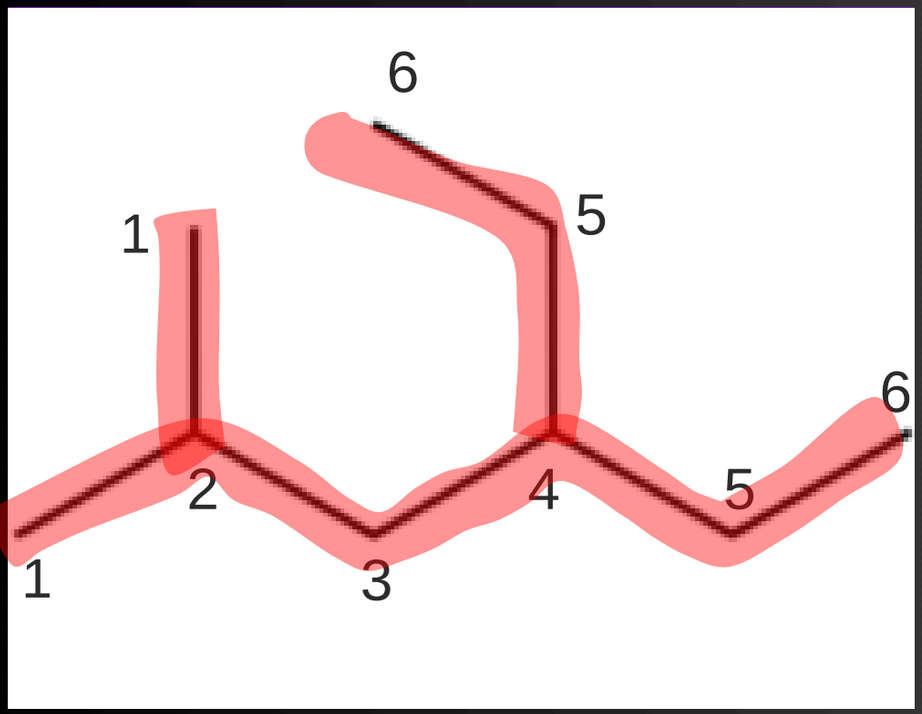
Parent: The longest chain is 4 C. There are multiple paths for the parent chain and all of it works since the compound is symmetrical
Numbering: Numbering here doesn't matter since the 2 methyl groups will always be at the 2nd and 3rd carbon.
Substituents: Two methyl groups at C2 and C3. 2,3-dimethyl

2,3-dimethylbutane



LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

4-ETHYL-2-METHYLHEXANE A04-DEVELOPING C_9H_{20}

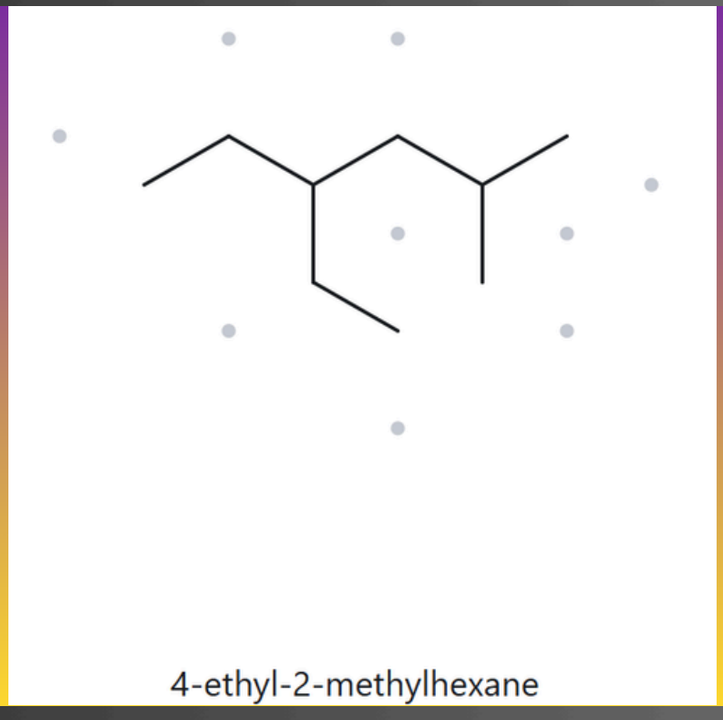


Parent: The longest chain is 6 C, so the parent is hexane. Several ways to obtain a 6C parent chain but not all work due to first point difference.

Numbering: Starting from the left gives you the lowest locant for a methyl group (substituent) and naming is based alphabetically E then M.

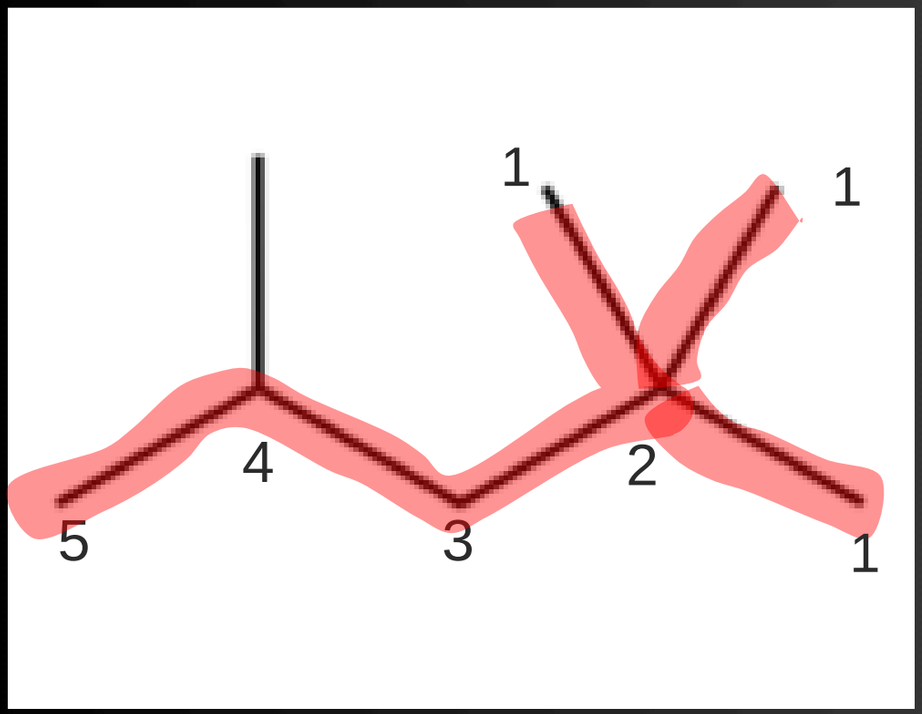
Substituents: One methyl group on C2 and one ethyl group on C4. 4-ethyl-2-methyl

4-ethyl-2-methylhexane



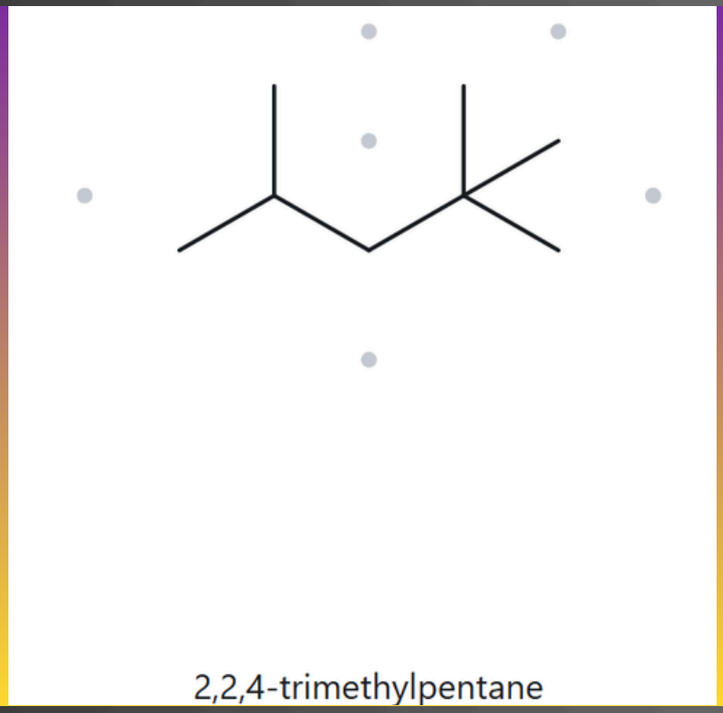
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

2,2,4-TRIMETHYLPENTANE AO5-DEVELOPING C_8H_{18}



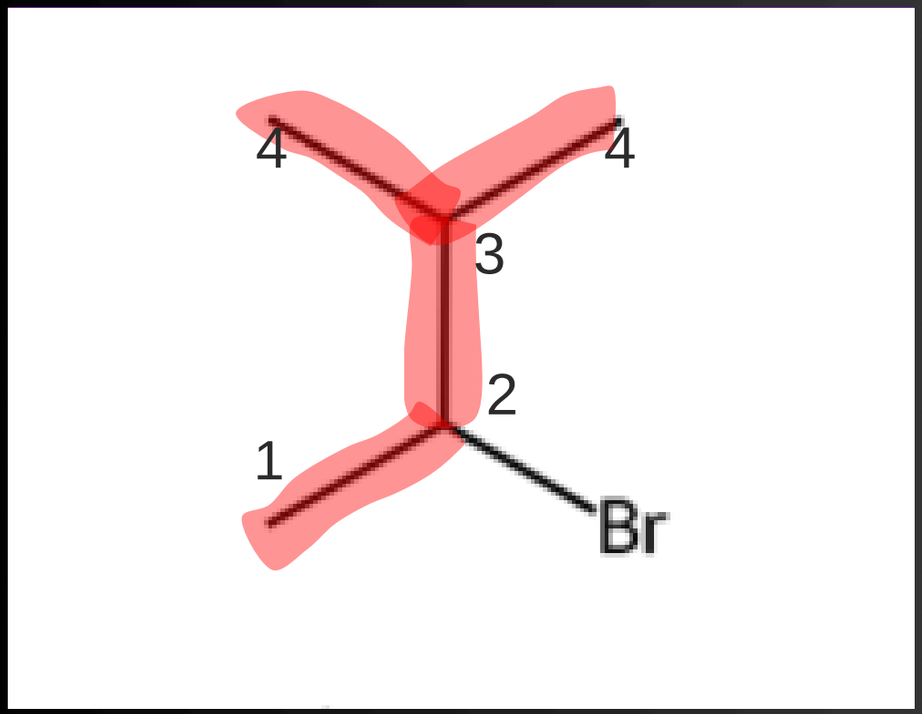
Parent: The longest chain is 5 C. There are several ways to obtain a 5 C parent chain. But only 3 works according to first point difference
Numbering: Starting from the right gives you the lowest locant for the two methyl groups attached to the same carbon.
Substituents: Three methyl groups, 2 in C2 and 1 in C4. 2,2,4-trimethyl

2,2,4-trimethylpentane



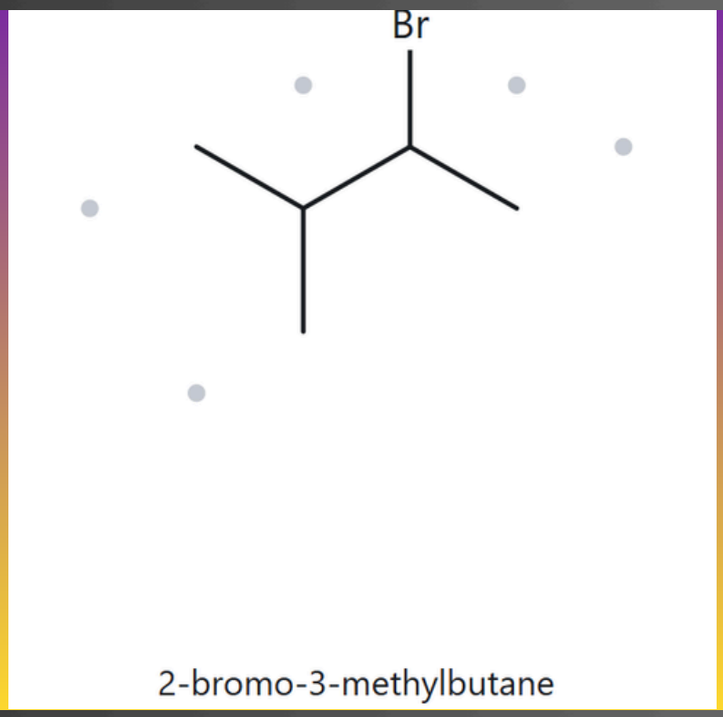
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

2-BROMO-3-METHYLBUTANE A06-DEVELOPING $C_5H_{11}BR$



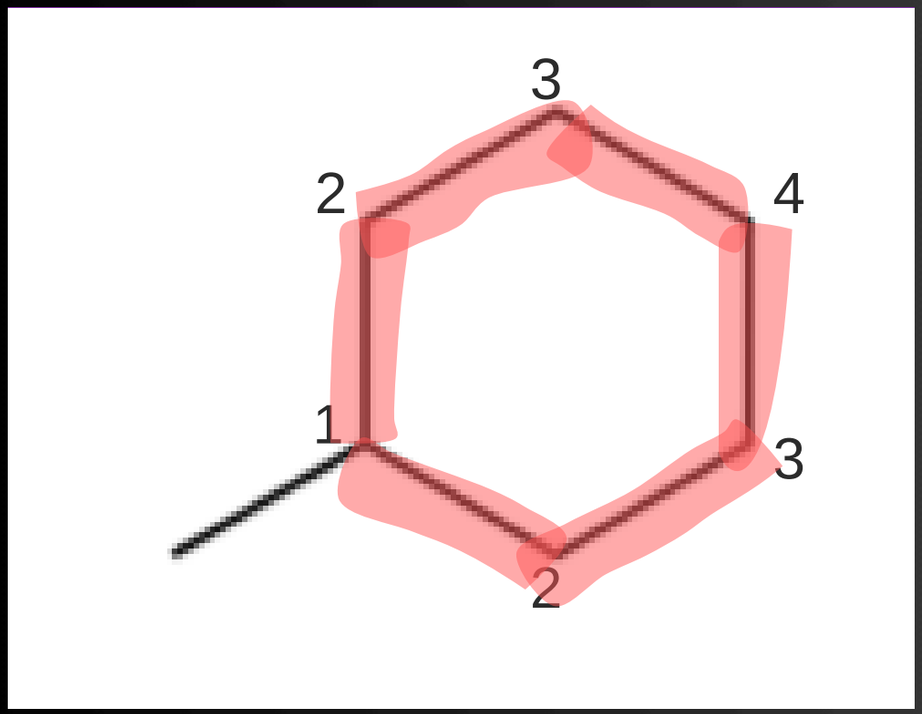
Parent: The longest chain is 4 C. Several wyas to obtain it but not all works due to tie in locants.
Numbering: Since there is a tie bromo group is given the lowest locant because alphabetically B comes first before M.
Substituents: One bromo group on C2 and one methyl group on C3. 2-bromo-3-methyl

2-bromo-3-methylbutane



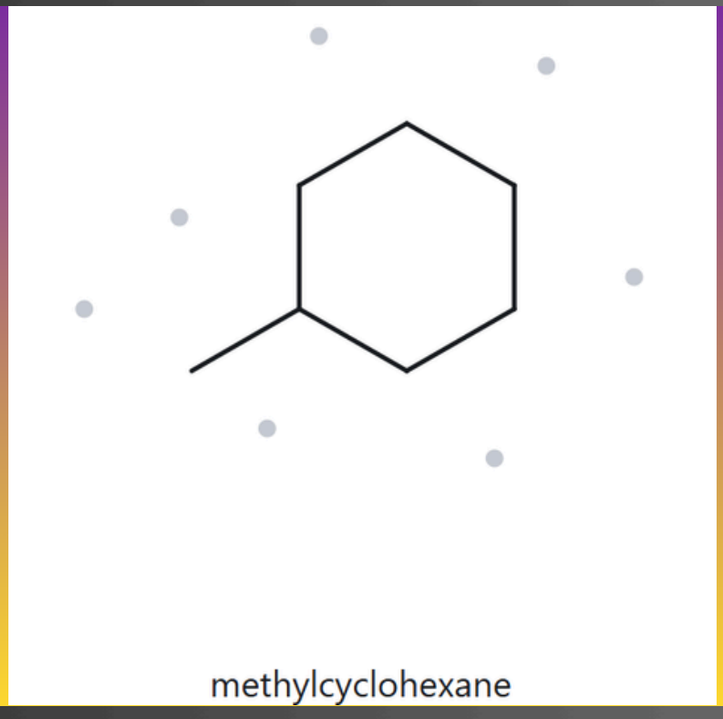
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

METHYLCYCLOHEXANE A07-DEVELOPING C_7H_{14}



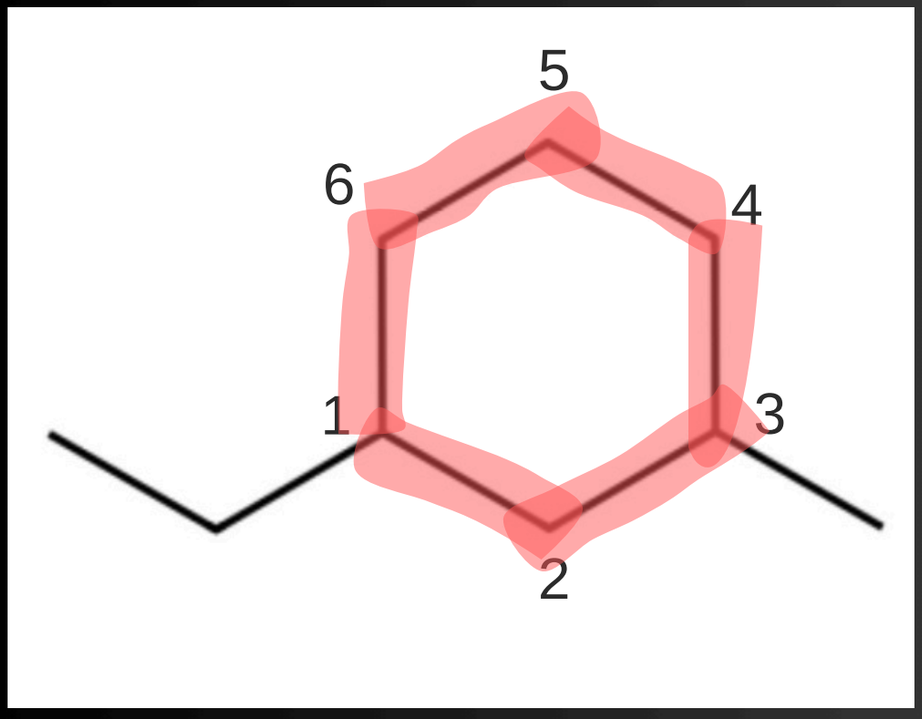
Parent: The longest chain is a 6-membered carbon ring (cyclohexane).
Numbering: Since there is only a single substituent on the ring, it is automatically assigned at locant 1.
Substituents: One methyl group on C1. 1-methly is unecessary so we write methyl directly.

methylcyclohexane



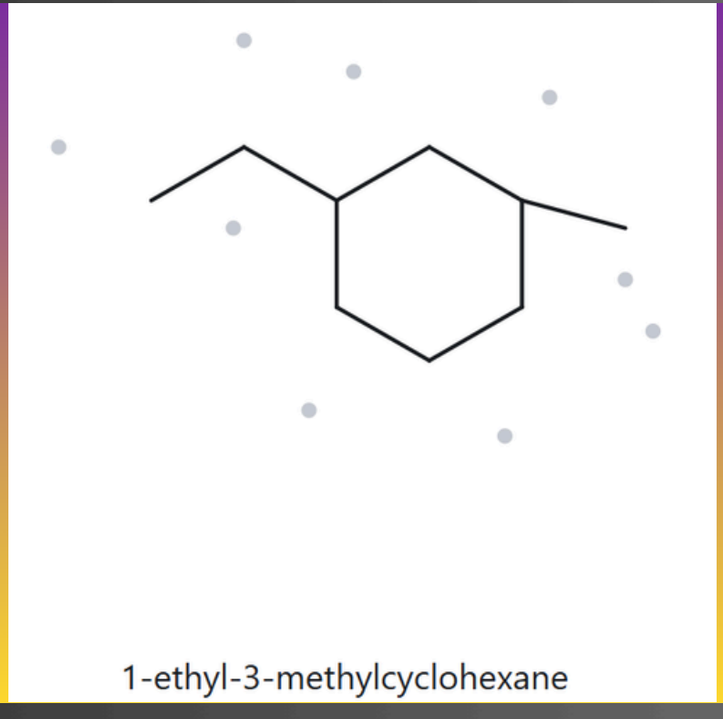
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

1-ETHYL-3-METHYLCYCLOHEXANE A08-CHALLENGE C_9H_{18}



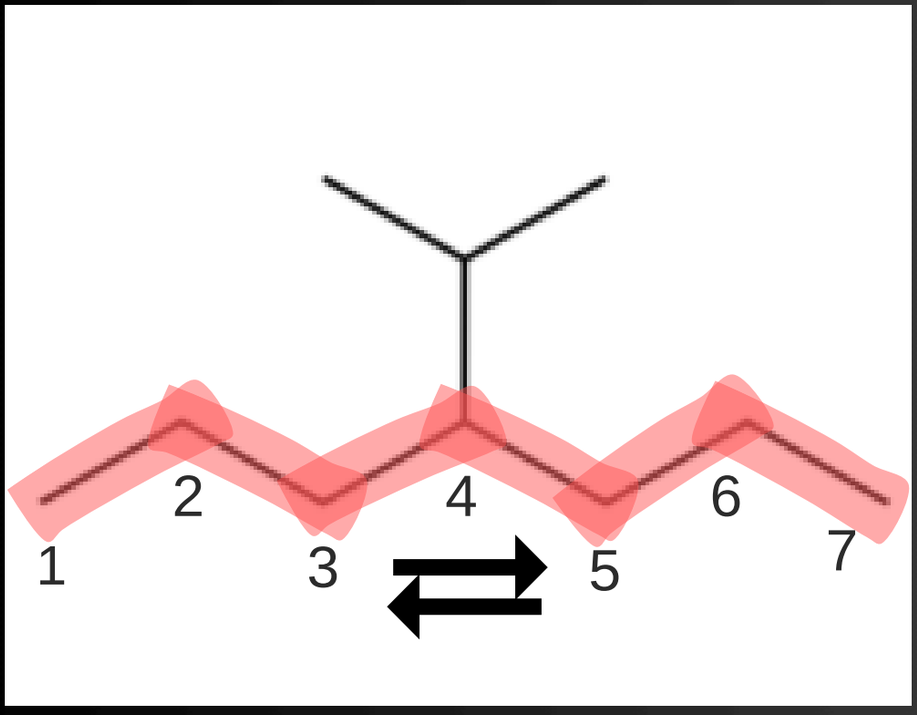
Parent: The longest chain is a 6-membered carbon ring (cyclohexane).
Numbering: Since there are two substituents and starting from either one results to locants 1 and 3 we break this tie alphabetically so ethyl gets lower locant.
Substituents: One ethyl group on C1 and one methyl group on C3.1-ethyl-3-methyl.

1-ethyl-3-methylcyclohexane



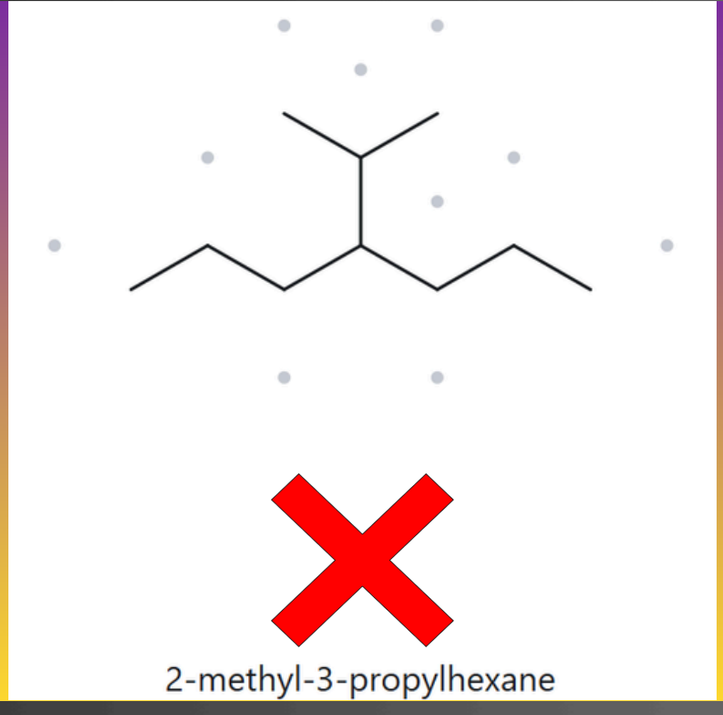
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

4-ISOPROPYLHEPTANE A09-CHALLENGE $C_{10}H_{22}$



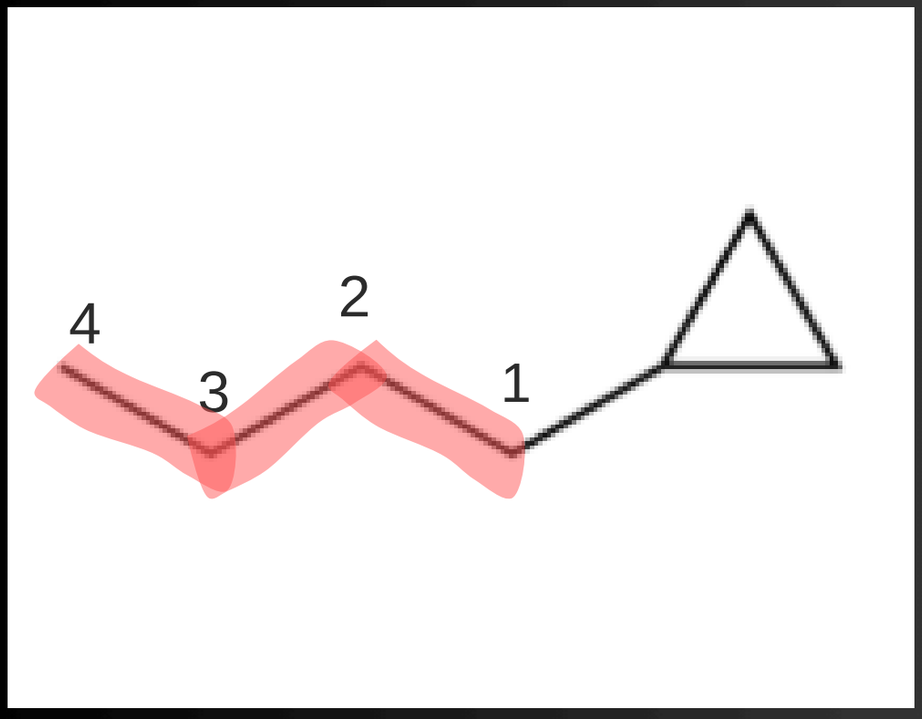
Parent: The longest chain is still composed of 7C unlike what thealkane.com says.
Numbering: Numbering can either be from the left or right of the 7C carbon chain since it is symmetrical.
Substituents: One propyl at C4 shaped like a Y making it isopropyl. 4-isopryl.

4-isopropylheptane



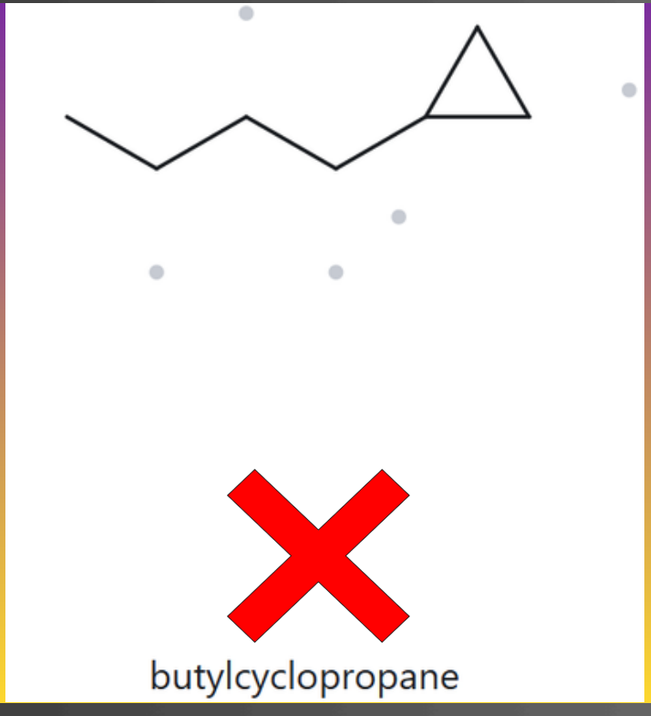
LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

CYCLOPROPYLBUTANE A10-CHALLENGE C_7H_{14}



Parent: The longest chain is still 4C chain since the attached carbon chain has more carbon than the cyclic (cyclopropyl).
Numbering: Numbering starts from the carbon attached with the cyclopropyl giving it a locant of 1.
Substituents: One cycloalkyl group on C1. 1-cyclopropyl, the 1 can be dropped but keeping it is preferred.

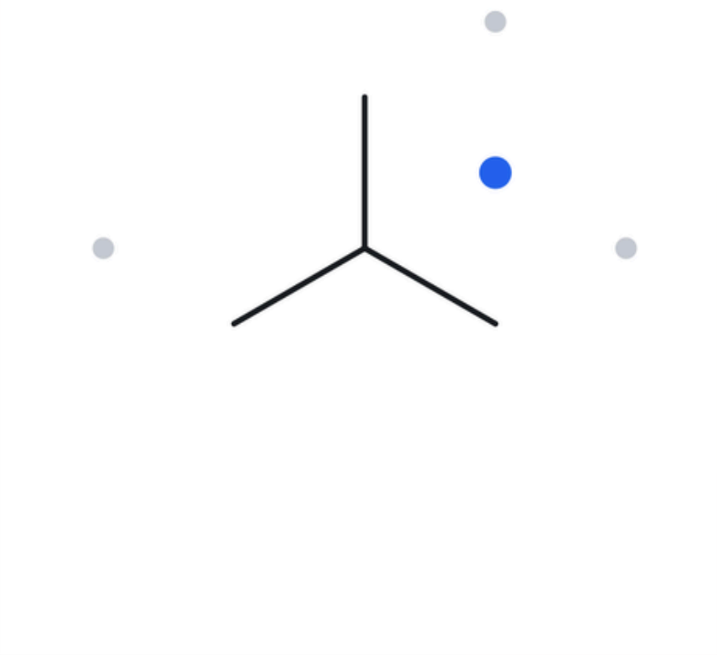
cyclopropylbutane



LONGEST CHAIN CONFIRMED✅/LOWEST LOCANTS✅/ALPHABETICAL ORDER✅/PUNCTUATION✅/FORMULA COUNTED✅

B01-FOUNDATION 2-METHYLPROPANE

Parent: propane, a 3-carbon chain
Substituents: methyl on C2
Expected Formula: 3+1= 4C; C_4H_{10}



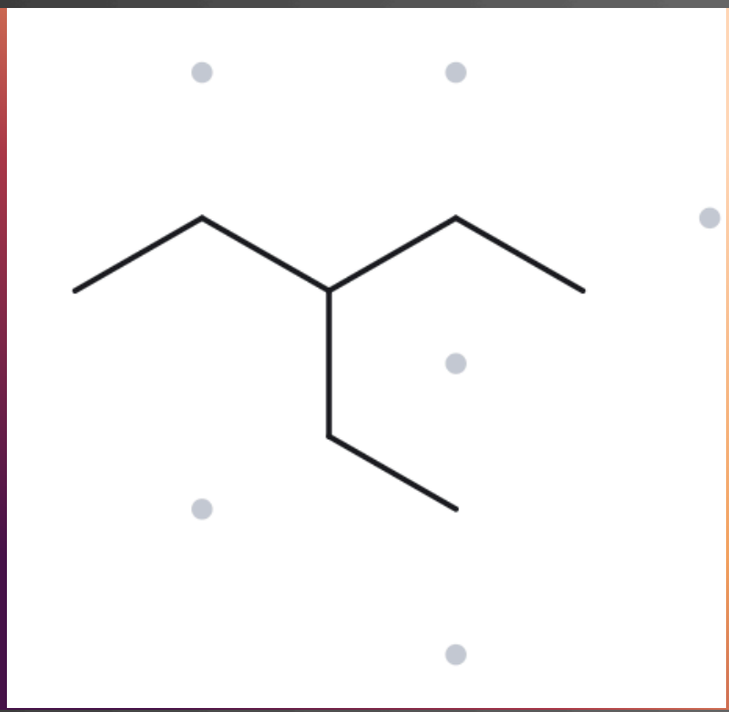
Counted from the drawing: C_4H_{10}
Alkane: C_nH_{2n+2} = C_4H_{10}

Returned by The Alkane: 2-methylpropane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B02-FOUNDATION 3-ETHYLPENTANE

Parent: pentane, a 5-carbon chain
Substituents: ethyl on C3
Expected Formula: 5+2= 7C; C_7H_{16}



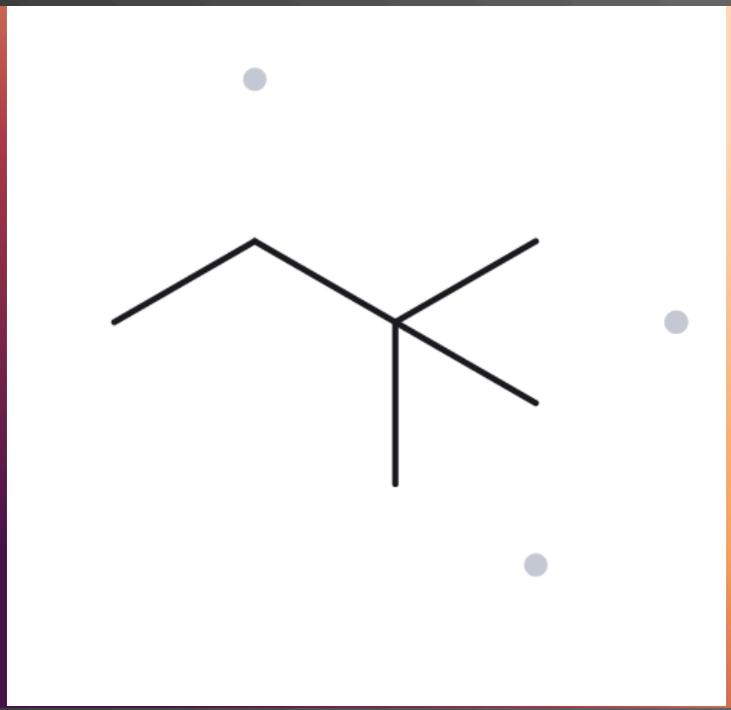
Counted from the drawing: C_7H_{16}
Alkane: $C_nH_{2n+2} = C_7H_{16}$

Returned by The Alkane: 3-ethylpentane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B03-FOUNDATION 2,2-DIMETHYLBUTANE

Parent: butane, a 4-carbon chain
Substituents: two methyl groups on C2
Expected Formula: 4+2= 6C; C_6H_{14}



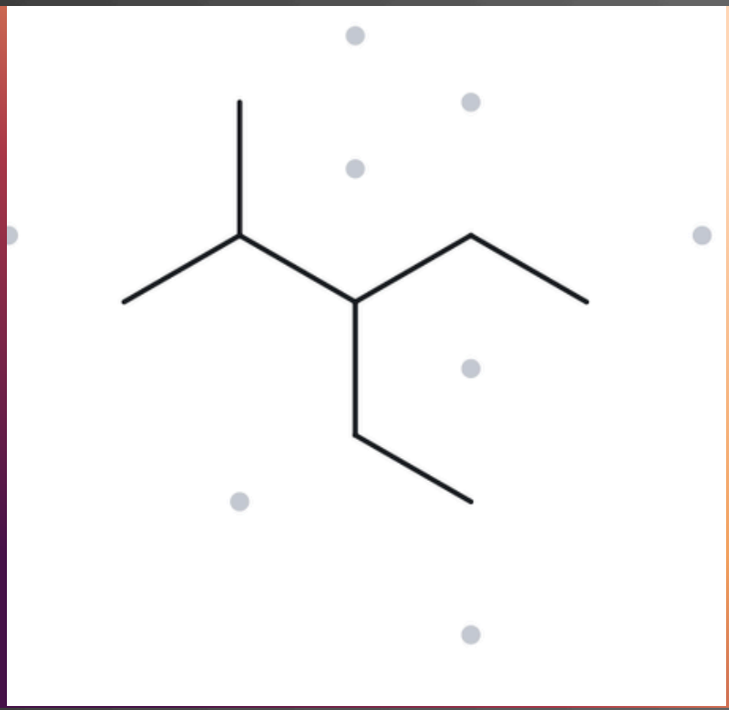
Counted from the drawing: C_6H_{14}
Alkane: $C_nH_{2n+2} = C_6H_{14}$

Returned by The Alkane: 2,2-dimethylbutane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B04-DEVELOPING 3-ETHYL-2-METHYLPENTANE

Parent: pentane, a 5-carbon chain
Substituents: methyl on C2 and ethyl on C3
Expected Formula: 5+1+2= 8C; C_8H_{18}



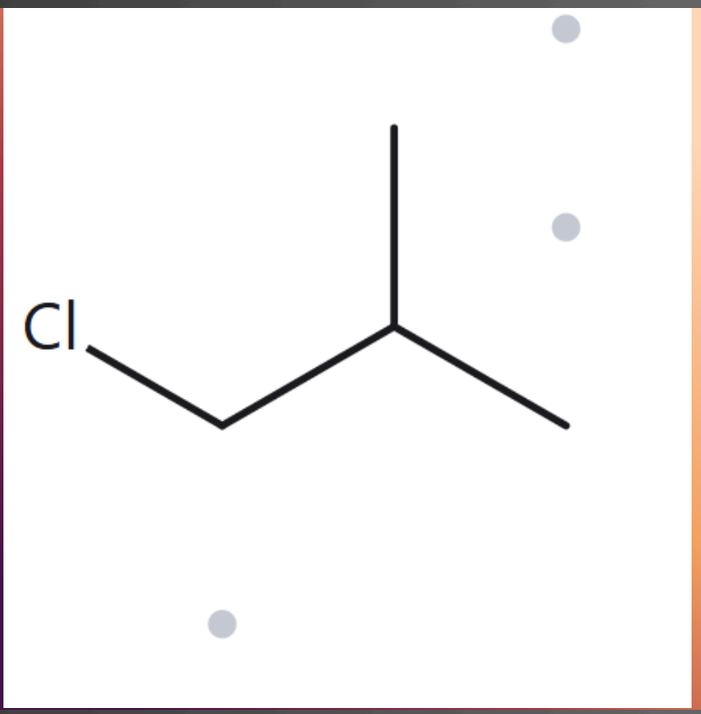
Counted from the drawing: C_8H_{18}
Alkane: C_nH_{2n+2} = C_8H_{18}

Returned by The Alkane: 3-ethyl-2-methylpentane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B05-DEVELOPING 1-CHLORO-2-METHYLPROPANE

Parent: propane, a 3-carbon chain
Substituents: chloro on C1 and methyl on C2
Expected Formula: 3+1= 4C; **C_4H_{10}; Cl replaces one H; C_4H_9Cl**



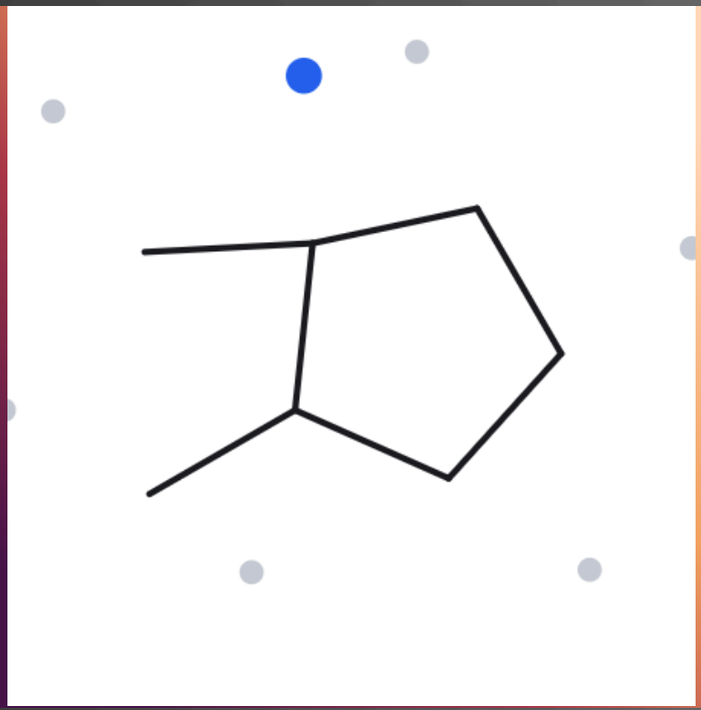
Counted from the drawing: C_4H_9Cl
Alkane: C_nH_{2n+2} = C_4H_{10} minus one H for Cl; C_4H_9Cl

Returned by The Alkane: 1-chloro-2-methylpropane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B06-DEVELOPING 1,2-DIMETHYLCYCLOPENTANE

Parent: cyclopentane, a 5-carbon ring
Substituents: two methyl groups on C1 and C2
Expected Formula: 5+2= 5C; **C_7H_{14}**



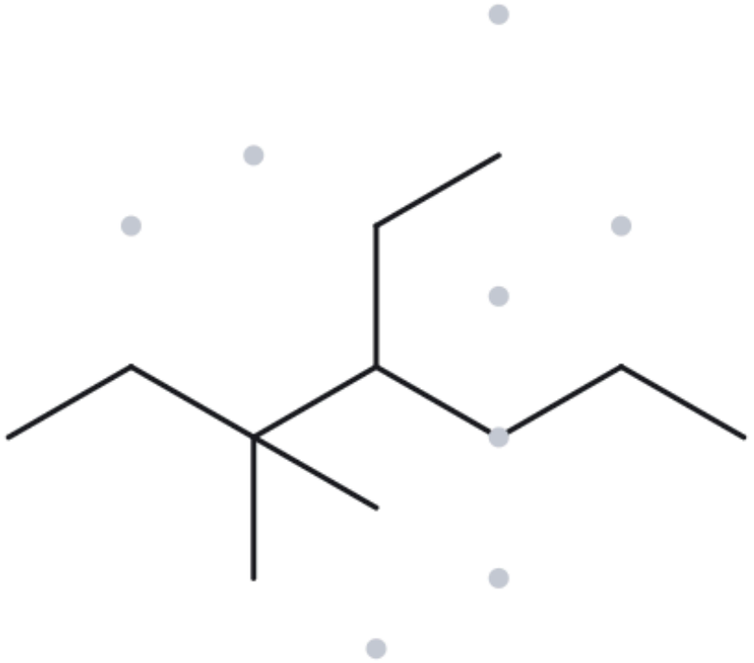
Counted from the drawing: C_7H_{14}
One ring: C_nH_{2n} = C_7H_{14}

Returned by The Alkane: 1,2-dimethylcyclopentane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B07-DEVELOPING 4-ETHYL-3,3-DIMETHYLHEPTANE

Parent: heptane, a 7-carbon chain
Substituents: one ethyl on C4 and two methyl groups on C3
Expected Formula: 7+2+2= 11C; $\mathbf{C_{11}H_{24}}$



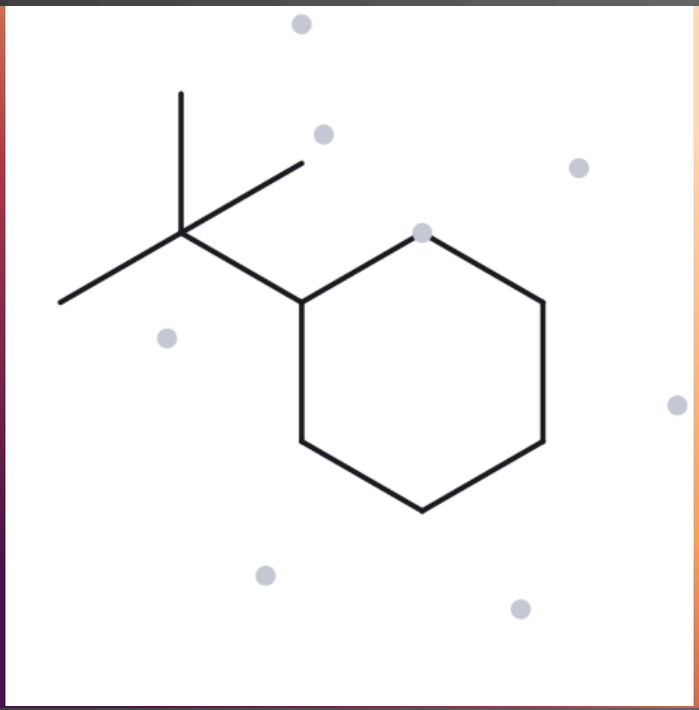
Counted from the drawing: $\mathbf{C_{11}H_{24}}$
Alkane: $\mathbf{C_nH_{2n+2}= C_{11}H_{24}}$

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B08-CHALLENGE TERT-BUTYLCYCLOHEXANE

Parent: cyclohexane, a 6-carbon ring
Substituents: tert-butyl on C1, any point in the ring can be C1 depending where the substituent is attached.
Expected Formula: 6+4= 10C; $\mathbf{C_{10}H_{20}}$



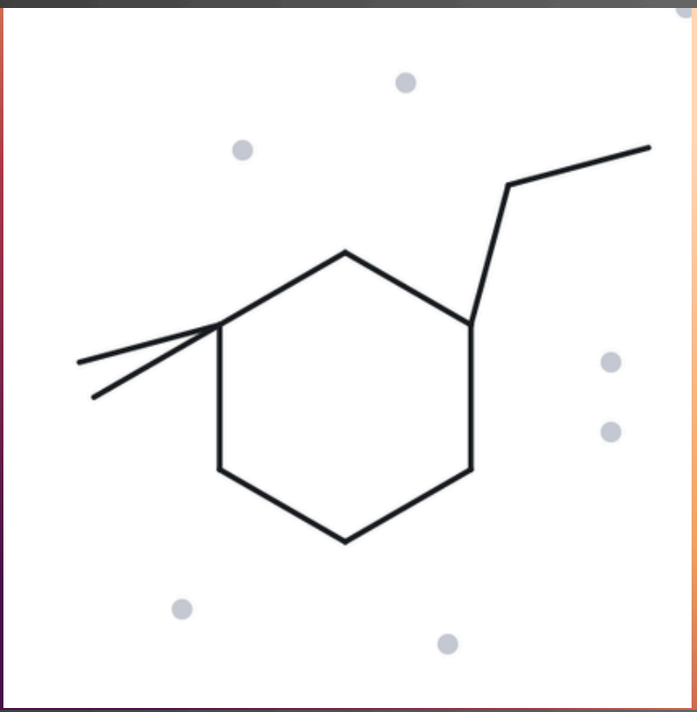
Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane
✅**Not literally same name but 1,1-dimethylethyl is commonly known as tert-butyl**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B09-CHALLENGE 3-ETHYL-1,1-DIMETHYLCYCLOHEXANE

Parent: cyclohexane, a 6-carbon ring
Substituents: ethyl on C3 and two methyl groups on C1
Expected Formula: 6+2+2= 10C; **$C_{10}H_{20}$**



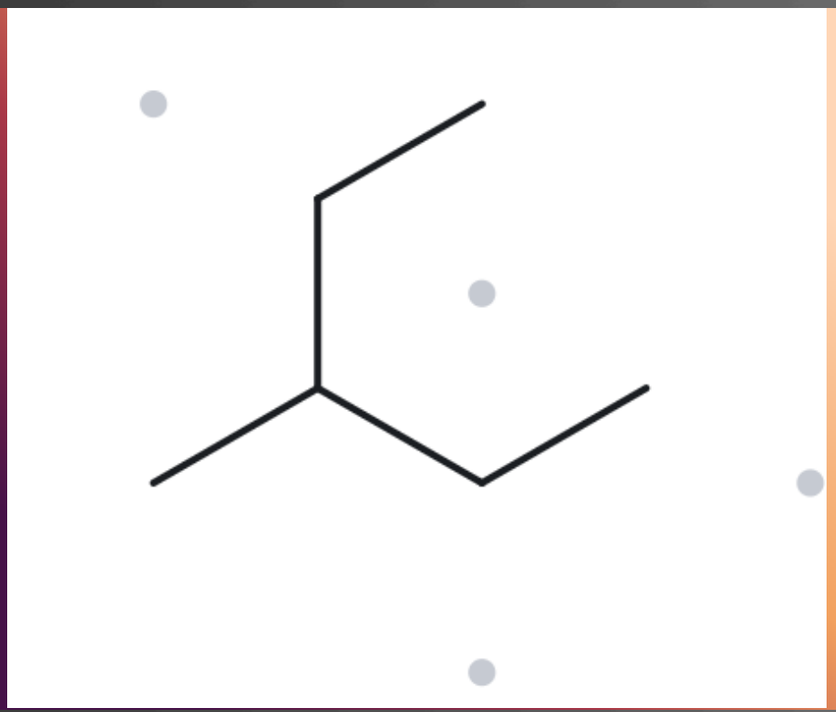
Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane
✅**Same as the given name**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅

B10-CHALLENGE 2-ETHYLBUTANE

Parent: butane, a 4-carbon chain
Substituents: one ethyl on C2
Expected Formula: 4+2= 6C; **C_6H_{14}**



Counted from the drawing: C_6H_{14}
Alkane: C_nH_{2n+2}= C_6H_{14}

Returned by The Alkane: 3-methylpentane
❌**The given name is incorrect since the longest carbon chain is 5**

PARENT LENGTH MATCHES✅/SUBSTITUENT POSITIONS MATCH✅/BACK-CHECK RECORDED✅/FORMULA AGREES✅